MATH1101: Differential Calculus & Analytical Geometry

Complete Exam Notes · 1st Year 1st Semester · PUST · Module 5 (Units 14 & 15): 2D Coordinates, Transformation of Coordinates & Pair of Straight Lines

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Module 5 Overview

Units Covered in Module 5:

- **Unit 14:** Two-Dimensional Coordinate Systems (Cartesian & Polar Coordinates, Parametric Equations) and Transformation of Coordinates (Translation of Origin, Rotation of Axes & Orthogonal Invariants).
- **Unit 15:** Pair of Straight Lines in 2D (Homogeneous Equation $ax^2 + 2hxy + by^2 = 0$, Angle & Bisectors, General Second-Degree Equation $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$, Homogenization & Solved PUST Past Exam Problems).

Master Roadmap (MATH 1101)

Mod.	Units & Core Syllabus Topics	Main Sources	Status
1	Units 1–3: Functions, Domain/Range, Graphs, Inverse, Limits & Continuity	Class Notes · Anton Ch. 0, 1 · PUST 20–23	Module 1
2	Units 4–5: Differentiability, Geometric Meaning, Extrema, y_n & Leibniz	Class Notes · Anton Ch. 2, 4 · PUST 20–23	Module 2
3	Units 6–9: Rolle's & MVT, Taylor/Maclaurin, L'Hôpital & Curve Tracing	Class Notes · Anton Ch. 3, 4, 9 · Handout	Module 3
4	Units 10–13: Tangents, Asymptotes, Curvature, Partial Deriv., Euler & Jacobian	Akib Notes · Anton Ch. 13 · PUST 20–23	Module 4
5	Units 14–15: 2D Coordinates, Transformation of Axes & Pair of Straight Lines	Anton Ch. 10 · Rahman & Bhattacharjee	✓ **This file**
6	Units 16–18: Conic Reduction, Parabola, Ellipse, Hyperbola & System of Circles	Anton Ch. 10 · Rahman & Bhattacharjee	Module 6
7	Units 19–21: 3D Coordinates, DCs/DRs, Projections, Planes & Skew Lines (S.D.)	Anton Ch. 11 · Rahman & Bhattacharjee	Module 7
8	Units 22–24: Sphere, Cone, Right Circular Cylinder & 3D Central Conicoids	Anton Ch. 11 · Rahman & Bhattacharjee	Module 8

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1 Unit 14: 2D Coordinate Systems and Transformation of Coordinates

1.1 Cartesian, Polar, and Parametric Equations

1. **Cartesian (x, y) and Polar (r, θ) Conversion:**

$$x = r\cos\theta, \qquad y = r\sin\theta,$$
$$r^2 = x^2 + y^2, \qquad \tan\theta = \frac{y}{x}.$$

2. **Distance Between $P_1(r_1, \theta_1)$ and $P_2(r_2, \theta_2)$ in Polar Coordinates:**

$$d = \sqrt{r_1^2 + r_2^2 - 2r_1 r_2 \cos(\theta_1 - \theta_2)}.$$

3. **Standard Parametric Equations (Anton Sec. 10.1):**

- *Circle* $x^2 + y^2 = a^2$: $x = a\cos t,\ y = a\sin t\ (0 \le t < 2\pi)$.
- *Parabola* $y^2 = 4ax$: $x = at^2,\ y = 2at\ (t \in \mathbb{R})$.
- *Ellipse* $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} = 1$: $x = a\cos\theta,\ y = b\sin\theta$.
- *Hyperbola* $\dfrac{x^2}{a^2} - \dfrac{y^2}{b^2} = 1$: $x = a\sec\theta,\ y = b\tan\theta$.

1.2 Transformation of Coordinates: Translation and Rotation

1. **Translation of Origin to (h, k) (Axes Parallel):**

$$x = x' + h, \qquad y = y' + k \qquad \Longleftrightarrow \qquad x' = x - h, \qquad y' = y - k.$$

To eliminate the first-degree terms $(2gx, 2fy)$ from $F(x, y) = ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$, the new origin (α, β) is found by solving:

$$\frac{1}{2}\frac{\partial F}{\partial x} = ax + hy + g = 0,$$
$$\frac{1}{2}\frac{\partial F}{\partial y} = hx + by + f = 0,$$

and the transformed equation becomes $aX^2 + 2hXY + bY^2 + c' = 0$, where $c' = g\alpha + f\beta + c$.

2. **Rotation of Axes Through Angle θ (Origin Fixed):**

$$x = x'\cos\theta - y'\sin\theta, \qquad x' = x\cos\theta + y\sin\theta,$$
$$y = x'\sin\theta + y'\cos\theta, \qquad y' = -x\sin\theta + y\cos\theta.$$

Important Note: Eliminating the xy-Term & Orthogonal Invariants

1. **Angle θ to Remove the xy-Term:**

$$\tan 2\theta = \frac{2h}{a - b} \qquad \text{or} \qquad \cot 2\theta = \frac{a - b}{2h} \qquad (\text{if } a = b, \text{ then } \theta = 45^\circ).$$

2. **Invariants:** Under rotation of axes, $a' + b' = a + b$ and $a'b' - h'^2 = ab - h^2$. When $h' = 0$, the new coefficients a', b' are the roots of the characteristic quadratic:

$$\lambda^2 - (a + b)\lambda + (ab - h^2) = 0.$$

1.3 Solved PUST Past Exam Problems on Transformation of Coordinates

PUST Past Exam Question 1 (Translation & Rotation of Axes) [7 Marks]

Question: Transform the equation $17x^2 + 18xy - 7y^2 - 16x - 32y - 18 = 0$ to one in which there are no first-degree terms and no xy-term.

Solution (Exam Style): Comparing with $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$:

$$a = 17, \quad h = 9, \quad b = -7, \quad g = -8, \quad f = -16, \quad c = -18.$$

Step 1 (Translation of Origin to (α, β) to Remove First-Degree Terms):

$$17\alpha + 9\beta - 8 = 0 \quad [\text{From } \frac{1}{2}\frac{\partial F}{\partial x} = 0]$$
$$9\alpha - 7\beta - 16 = 0 \quad [\text{From } \frac{1}{2}\frac{\partial F}{\partial y} = 0]$$
$$\implies \alpha = 1, \qquad \beta = -1.$$

Computing the new constant term c':

$$c' = g\alpha + f\beta + c$$
$$= (-8)(1) + (-16)(-1) - 18 = -10.$$

Thus, shifting the origin to $(1, -1)$ gives:

$$17X^2 + 18XY - 7Y^2 - 10 = 0.$$

Step 2 (Rotation of Axes to Remove the XY-Term): By orthogonal invariants:

$$a' + b' = a + b = 17 - 7 = 10$$
$$a'b' = ab - h^2 = (17)(-7) - 9^2 = -200$$

Thus a', b' are the roots of:

$$\lambda^2 - 10\lambda - 200 = 0$$
$$\implies (\lambda - 20)(\lambda + 10) = 0$$
$$\implies \lambda = 20, -10.$$

Substituting $a' = 20$ and $b' = -10$ into $a'x'^2 + b'y'^2 + c' = 0$:

$$20x'^2 - 10y'^2 - 10 = 0$$
$$\implies 2x'^2 - y'^2 = 1.$$

Problem 14.1 (PUST Past Exam): Transform $x^2 + 4xy + y^2 - 2x + 2y - 6 = 0$ to standard form.

Solution (Exam Style): Here $a = 1, h = 2, b = 1, g = -1, f = 1, c = -6$.

Center equations: $\alpha + 2\beta - 1 = 0$ and $2\alpha + \beta + 1 = 0$

$$\implies \alpha = -1, \qquad \beta = 1$$

New constant: $c' = g\alpha + f\beta + c = (-1)(-1) + (1)(1) - 6 = -4$

Characteristic equation: $\lambda^2 - (1+1)\lambda + (1-4) = 0$

$$\implies \lambda^2 - 2\lambda - 3 = 0 \implies \lambda = 3, -1$$

$$\therefore 3x'^2 - y'^2 - 4 = 0 \implies \frac{x'^2}{4/3} - \frac{y'^2}{4} = 1.$$

2 Unit 15: Pair of Straight Lines in 2D

2.1 Homogeneous Second-Degree Equation: $ax^2 + 2hxy + by^2 = 0$

The homogeneous equation $ax^2 + 2hxy + by^2 = 0$ ($h^2 \geq ab$) represents two straight lines passing through the origin $(0, 0)$, $y = m_1 x$ and $y = m_2 x$, where:

$$m_1 + m_2 = -\frac{2h}{b}, \qquad m_1 m_2 = \frac{a}{b}.$$

1. **Theorem (Angle θ Between the Pair of Lines):**

$$\tan\theta = \left|\frac{m_1 - m_2}{1 + m_1 m_2}\right| = \frac{\sqrt{(m_1 + m_2)^2 - 4m_1 m_2}}{|1 + m_1 m_2|}$$
$$= \frac{\sqrt{\frac{4h^2}{b^2} - \frac{4a}{b}}}{\left|1 + \frac{a}{b}\right|} = \frac{2\sqrt{h^2 - ab}}{|a + b|}.$$

   - **Perpendicular Lines ($\theta = 90°$):** $a + b = 0$ (coeff. of x^2 + coeff. of $y^2 = 0$).
   - **Coincident / Parallel Lines ($\theta = 0°$):** $h^2 = ab$.

2. **Theorem (Combined Equation of Angle Bisectors):**

$$\frac{x^2 - y^2}{a - b} = \frac{xy}{h} \qquad \Longleftrightarrow \qquad hx^2 - (a - b)xy - hy^2 = 0.$$

2.2 General Second-Degree Equation: $S \equiv ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$

1. **Condition to Represent a Pair of Straight Lines:**

$$\Delta = \begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix} = abc + 2fgh - af^2 - bg^2 - ch^2 = 0 \qquad (\text{with } h^2 \geq ab).$$

2. **Point of Intersection:** $(x_0, y_0) = \left(\frac{hf - bg}{ab - h^2}, \frac{gh - af}{ab - h^2}\right)$, found by solving $\frac{\partial S}{\partial x} = 0$ and $\frac{\partial S}{\partial y} = 0$.

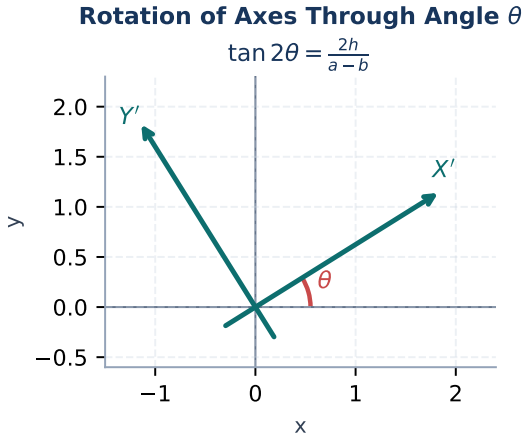
3. **Distance Between Parallel Lines (when $h^2 = ab$):** $d = 2\sqrt{\frac{g^2 - ac}{a(a + b)}} = 2\sqrt{\frac{f^2 - bc}{b(a + b)}}$.

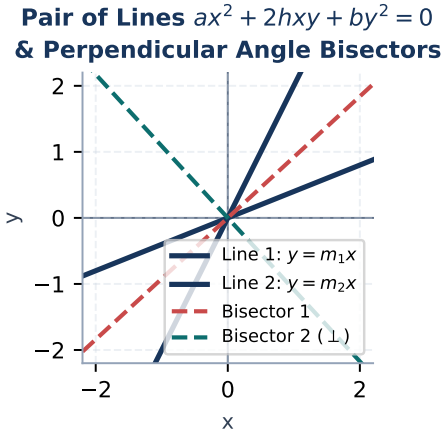
2.3 Homogenization: Lines Joining the Origin to the Intersection of a Curve and a Line

To find the pair of straight lines joining $(0, 0)$ to the intersection of $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ and $lx + my + n = 0$:

Step 1: $\frac{lx + my}{-n} = 1$ [Isolating 1 from the linear equation]

Step 2: $ax^2 + 2hxy + by^2 + (2gx + 2fy)\left(\frac{lx + my}{-n}\right) + c\left(\frac{lx + my}{-n}\right)^2 = 0$ [Homogeneous of degree 2].





2.4 Solved PUST Past Exam Problems on Pair of Straight Lines

PUST Past Exam Question 1 (General Pair of Straight Lines) [5 Marks]

Question: Prove that $2x^2 + 7xy + 3y^2 + 8x + 14y + 8 = 0$ represents a pair of straight lines. Find their separate equations, point of intersection, and the angle between them.

Solution (Exam Style): Here, $a = 2,\ h = \frac{7}{2},\ b = 3,\ g = 4,\ f = 7,\ c = 8.$

Step 1 (Testing $\Delta = 0$):

$$\begin{aligned}\Delta &= abc + 2fgh - af^2 - bg^2 - ch^2 \\ &= (2)(3)(8) + 2(7)(4)\left(\frac{7}{2}\right) - 2(49) - 3(16) - 8\left(\frac{49}{4}\right) \\ &= 48 + 196 - 98 - 48 - 98 = 0 \quad [\text{Also } h^2 - ab = \frac{49}{4} - 6 = \frac{25}{4} > 0].\end{aligned}$$

Hence the equation represents a pair of real straight lines.

Step 2 (Separate Equations & Point of Intersection):

$$2x^2 + 7xy + 3y^2 = (2x + y)(x + 3y)$$

Let $2x^2 + 7xy + 3y^2 + 8x + 14y + 8 \equiv (2x + y + c_1)(x + 3y + c_2)$

$$\implies c_1 + 2c_2 = 8 \quad \text{and} \quad 3c_1 + c_2 = 14 \implies c_1 = 4, \quad c_2 = 2.$$

Thus the two separate lines are $2x + y + 4 = 0$ and $x + 3y + 2 = 0$, which intersect at $(-2, 0)$.

Step 3 (Angle Between the Lines):

$$\begin{aligned}\tan\theta &= \frac{2\sqrt{h^2 - ab}}{a + b} \\ &= \frac{2\sqrt{\frac{49}{4} - 6}}{2 + 3} = \frac{5}{5} = 1 \\ &\implies \theta = 45^\circ.\end{aligned}$$

PUST Past Exam Question 2 (Homogenization Proof) [5 Marks]

Question: Show that the straight lines joining the origin to the points of intersection of the circle $x^2 + y^2 = a^2$ and the line $lx + my = 1$ are at right angles if $a^2(l^2 + m^2) = 2$.

Solution (Exam Style):

Given circle: $x^2 + y^2 = a^2$

Given line: $lx + my = 1$

Homogenizing the circle with $lx + my = 1$: $x^2 + y^2 = a^2(1)^2$

$\implies x^2 + y^2 = a^2(lx + my)^2$

$\implies x^2 + y^2 = a^2(l^2x^2 + 2lm\,xy + m^2y^2)$

$\implies (1 - a^2l^2)x^2 - 2a^2lm\,xy + (1 - a^2m^2)y^2 = 0$ [Pair of lines through $(0, 0)$].

The two lines are perpendicular if coeff. of x^2 + coeff. of $y^2 = 0$:

$(1 - a^2l^2) + (1 - a^2m^2) = 0$

$\implies a^2(l^2 + m^2) = 2.$ ■

Quick Revision — Module 5 (Units 14–15)

• **Translation & Rotation:** Center (α, β) from $\frac{\partial F}{\partial x} = 0, \frac{\partial F}{\partial y} = 0$; $c' = g\alpha + f\beta + c$; $\tan 2\theta = \frac{2h}{a-b}$; $\lambda^2 - (a + b)\lambda + (ab - h^2) = 0$.

• **Pair of Lines:** $\Delta = abc + 2fgh - af^2 - bg^2 - ch^2 = 0$; $\tan\theta = \frac{2\sqrt{h^2-ab}}{|a+b|}$ ($\perp \iff a + b = 0$); Bisectors: $\frac{x^2-y^2}{a-b} = \frac{xy}{h}$.

• **Homogenization:** Isolate $1 = \frac{lx+my}{-n}$ from the line and multiply degree-1 terms by $\left(\frac{lx+my}{-n}\right)$ and constant c by $\left(\frac{lx+my}{-n}\right)^2$.

Common Mistakes to Avoid in the Exam

• Using $2g, 2f, 2h$ instead of g, f, h in $\Delta = abc + 2fgh - af^2 - bg^2 - ch^2 = 0$ and $c' = g\alpha + f\beta + c$: always halve the coefficients of xy, x, y first!